

NUR205 Medical-Surgical Nursing IV

EXAM 1 INTEGRATED STUDY GUIDE

Endocrine, Diabetes Mellitus, and Kidney: what each one is, and how they connect

Chapter (verified by title in your book)	Content
Chapter 53	Assessment of the Endocrine System
Chapter 54	Pituitary and Adrenal Gland Conditions
Chapter 55	Thyroid and Parathyroid Gland Conditions
Chapter 56	Diabetes Mellitus, including diabetic ketoacidosis and hyperglycemic-hyperosmolar state
Chapter 60	Acute Kidney Injury and Chronic Kidney Disease, including dialysis and kidney transplant

Source: Ignatavicius, Medical-Surgical Nursing, 11th edition. Every lab value, drug point, and nursing action in this guide was checked against the chapter text. The "Like you're 10" boxes and the connection sections are plain-language explanations built on those chapter facts; they are teaching aids, not quotations.

Color key

Color	Meaning
Blue box	"Like you're 10": the simple version first
Red box	Safety priority or life-threatening content
Green box	Nursing interventions and client education
Orange box	Drugs
Purple box	Clinical judgment: what the exam is really asking
Teal box	Pearl: the fact that separates two look-alike answers
Gray box	Connection to another chapter
Gold box	Practice questions

Numbers versus bullets

Numbered steps (1, 2, 3) are nursing management in the order you do them. Where the chapter gives a sequence, the numbers follow it. Where it does not, the order is priority order: airway and breathing, then circulation, assess before you intervene, actual problems before potential ones.

Bullets have no order. They are used only for lists of cues, lab findings, drug facts, and teaching points.

How to use this in three days

- Read Part 1 first. It is the map. Every later chapter hangs on it.
- For each condition, read the blue box, then the table, then the red box. Stop there on a first pass.
- Work in 25-minute blocks with 5-minute breaks. One condition per block.
- Finish with Part 7 (connections) and Part 8 (self-test). Do not read the answer key until you have committed to every answer.

PART 1 | THE BIG PICTURE: ONE MAP FOR ALL FIVE CHAPTERS

Like you're 10

Your body is a house. The **endocrine glands** are the people who send notes (hormones) through the hallways (blood) telling rooms what to do. The **kidneys** are the plumbing and the trash service: they decide how much water and salt stays, and they take out the garbage.

Most of the notes in these chapters are **addressed to the kidney**. That is why endocrine and kidney are on the same exam. When a gland sends too many or too few notes, the plumbing floods or runs dry. When the plumbing itself breaks, the notes stop working.

The five things the body is trying to keep steady

Almost every question on this exam is one of these five balances going wrong. Learn which hormone runs each balance and which organ carries out the order.

Balance	Who controls it	Too little control	Too much control
Water	Vasopressin (antidiuretic hormone, ADH) tells the kidney to keep water	Diabetes insipidus: huge volumes of dilute urine, dehydration, sodium rises	Syndrome of inappropriate antidiuretic hormone: water kept, sodium diluted
Sodium and potassium	Aldosterone tells the kidney to keep sodium and water and excrete potassium	Adrenal insufficiency: sodium low, potassium high, blood pressure low	Hyperaldosteronism and Cushing syndrome: sodium high, potassium low, blood pressure high
Glucose	Insulin lowers it. Glucagon, cortisol, and other hormones raise it	No insulin effect: diabetes mellitus, diabetic ketoacidosis, hyperglycemic-hyperosmolar state	Too much insulin effect: hypoglycemia
Metabolic speed	Thyroid hormones (T3 and T4) set how fast every cell runs	Hypothyroidism: everything slow and cold. Extreme: myxedema coma	Hyperthyroidism: everything fast and hot. Extreme: thyroid storm
Calcium and phosphorus	Parathyroid hormone (PTH) raises calcium; calcitonin lowers it; the kidney activates vitamin D	Hypoparathyroidism: calcium low, tetany	Hyperparathyroidism: calcium high, bone loss, kidney stones

Where the kidney fits

The kidney is the **target organ** for vasopressin, aldosterone, and parathyroid hormone. It also activates vitamin D, makes erythropoietin, excretes potassium, phosphorus, and acid, and clears insulin.

So when the kidney fails (Chapter 60), you see the same lab problems you learned in the endocrine chapters, all at once: potassium high, phosphorus high, calcium low, parathyroid hormone high, metabolic acidosis, anemia, and insulin lasting longer than expected.

The chain of command

Like you're 10

Think of a company. The **hypothalamus** is the owner. The **pituitary** is the manager. The **thyroid and adrenal glands** are the workers. The owner tells the manager, the manager tells the worker, and the worker makes the product. When there is enough product, the worker's output tells the owner and manager to stop asking. That "stop asking" signal is **negative feedback**.

Owner (hypothalamus)	Manager (anterior pituitary)	Worker	Product
Corticotropin-releasing hormone (CRH)	Adrenocorticotrophic hormone (ACTH)	Adrenal cortex	Cortisol
Thyrotropin-releasing hormone (TRH)	Thyroid-stimulating hormone (TSH)	Thyroid	T3 and T4

Pearl: primary versus secondary

A **primary** problem is in the worker gland. A **secondary** problem is in the manager (pituitary).

- Worker broken, manager fine: the manager shouts louder. **Product low, stimulating hormone high**. Example: primary hypothyroidism has low T3 and T4 with **high TSH**.
- Manager broken: **product low, stimulating hormone low or near normal**. Example: secondary hypothyroidism.
- In primary adrenal insufficiency, ACTH is high. Melanocyte-stimulating hormone rises with it, which is why the skin **hyperpigments**. In secondary adrenal insufficiency, skin pigmentation does not change.

PART 2 | CHAPTER 53: ASSESSMENT OF THE ENDOCRINE SYSTEM

Like you're 10

A hormone is a key. It floats everywhere in the blood, but it only opens the doors that have its matching lock (a receptor). Cells with that lock are the **target tissue**. No lock, no effect.

Three things can go wrong: **too many keys**, **too few keys**, or **locks that do not work** (poor hormone-receptor interaction).

Glands and hormones you must know cold

Gland	Hormone	What it does
Posterior pituitary	Vasopressin (antidiuretic hormone)	Promotes water reabsorption in the kidney. Made in the hypothalamus, stored in the posterior pituitary
Posterior pituitary	Oxytocin	Uterine contractions and breast milk secretion
Anterior pituitary	ACTH, TSH, growth hormone, prolactin, luteinizing hormone, follicle-stimulating hormone, melanocyte-stimulating hormone	Mostly "manager" hormones that act on other glands. Growth hormone promotes growth through lipolysis, protein anabolism, and insulin antagonism
Adrenal cortex	Aldosterone (mineralocorticoid)	Sodium and water reabsorption, potassium excretion. Regulated by the renin-angiotensin-aldosterone system, serum potassium, and ACTH
Adrenal cortex	Cortisol (glucocorticoid)	Essential for life. Stress response; prevents hypoglycemia; anti-inflammatory; maintains cardiac muscle responsiveness. Peaks in the morning, lowest 12 hours after the peak
Adrenal medulla	Epinephrine (85%) and norepinephrine (15%)	Fight-or-flight. Not essential for life because other tissues also secrete them
Thyroid	T3, T4	Regulate metabolic rate and heat production; serve as insulin antagonists . Need dietary iodine and protein
Thyroid	Calcitonin	Lowers serum calcium and phosphorus. Opposite of parathyroid hormone
Parathyroid	Parathyroid hormone	Raises serum calcium: pulls calcium from bone, activates vitamin D in the kidney, reabsorbs calcium in the kidney, increases phosphorus excretion
Pancreas	Insulin (beta cells), glucagon (alpha cells), somatostatin (delta cells)	Insulin lowers glucose. Glucagon raises it. Somatostatin inhibits both

Connection: why steroids, thyroid disease, and acromegaly raise glucose

The chapter states that cortisol increases liver glucose production, that thyroid hormones are insulin antagonists, and that growth hormone works partly through insulin antagonism. That is the reason hyperglycemia shows up in Cushing syndrome, hyperthyroidism, and acromegaly, and hypoglycemia shows up in adrenal insufficiency and myxedema coma.

Age-related changes (frequently tested as "expected versus needs follow-up")

Change	Result	Nursing action
Decreased antidiuretic hormone	Dilute urine; cannot concentrate when intake is low; dehydration risk	Offer fluids at least every 2 hours while awake unless contraindicated
Decreased ovarian estrogen	Lower bone density; thin dry skin; drier perineal tissue with more cystitis	Weight-bearing exercise; protect skin; perineal care at least twice daily
Decreased glucose tolerance	Higher fasting and random glucose; slow wound healing; frequent yeast infections	Suggest diabetes testing for persistent vaginal candidiasis or a leg or foot wound not healed in 2 weeks
Decreased general metabolism	Less cold tolerance; decreased appetite; lower heart rate and blood pressure	Dress in layers. Follow up lethargy, new constipation, temperature consistently below 97 F, heart rate consistently below 60

Physical assessment

- **Inspection:** prominent forehead or jaw, round or puffy face, exophthalmos, visible thyroid enlargement, hyperpigmentation or vitiligo, striae, buffalo hump with truncal obesity, hirsutism, gynecomastia.
- **Thyroid:** normally not visible. Palpate standing **behind** the client, head bent slightly forward, and have the client swallow. Auscultate an enlarged thyroid for **bruits**.
- **Auscultation:** heart rate and rhythm; orthostatic vital signs, because some endocrine problems cause dysrhythmias and others cause volume depletion.

Safety priority

Do not palpate or apply pressure to the thyroid of a client who has or may have hyperthyroidism. It can release a sudden surge of thyroid hormone and trigger thyroid storm.

Diagnostic testing

Test	When it is used	How to read it
Stimulation (provocative) test	Gland suspected of being underactive	Hormone level fails to rise = hypofunction
Suppression test	Hormone levels high or high-normal	Hormone level fails to drop = hyperfunction
24-hour urine	Hormones released in pulses, such as adrenal hormones	Reflects gland function better than one blood sample
Glycosylated hemoglobin (A1C)	Average glucose over the life of the red blood cell	Not altered by what the client ate the day before

Client education: 24-hour urine collection, in order

1. Empty the bladder and **discard** that urine. Note the time. The clock starts now.
2. Save **all** urine from that point on. Keep the container in a cooler with ice if the specimen must stay cold; never store it with food or drinks.
3. At the end time, empty the bladder even without an urge and **add** that urine to the container.

Throughout: do not discard the preservative, and avoid splashing because some preservatives are caustic. Ask the provider before starting any new prescribed or over-the-counter drug during testing.

- **Blood draws:** catecholamine samples go on ice and straight to the laboratory. Do not draw hormone samples from a double- or triple-lumen line.
- **Imaging:** MRI with contrast is the most sensitive for the pituitary. CT is usually used for the adrenal glands. Ultrasound evaluates the thyroid and parathyroid glands.

PART 3 | CHAPTER 54: PITUITARY AND ADRENAL GLAND CONDITIONS

Priority concept: fluid and electrolyte balance. Read every condition in this chapter by asking two questions: where is the water going, and what is happening to sodium and potassium?

Hypopituitarism

Like you're 10

The manager stops sending orders. The worker glands are healthy, but nobody tells them to work, so their products run low.

- Most common cause: pituitary tumors. Also shock or severe hypotension, head trauma, radiation or surgery of the head, malnutrition.
- **Deficiency of ACTH or TSH is the most life-threatening**, because cortisol and thyroid hormone drop.
- Tumor growth often shows first as **vision changes**, especially peripheral vision; also headaches and diplopia.
- Treatment is **lifelong hormone replacement**, and it must continue during any hospital admission.

Drugs

- **Testosterone** for men with gonadotropin deficiency. Avoided in prostate cancer. Side effects: gynecomastia, acne, baldness, prostate enlargement.
- **Estrogen and progesterone** for women. Estrogen raises the risk for hypertension and thrombosis, especially with nicotine use.
- **Human growth hormone** subcutaneously **at night**, after the last meal, to mimic normal release.

Hyperpituitarism

Like you're 10

A lump in the manager's office (usually a benign adenoma) keeps shouting one order nonstop. The lump also takes up space in the skull, so it presses on its neighbors: the eyes and the brain.

Hormone in excess	What you see
Prolactin (most common adenoma)	Galactorrhea, amenorrhea, infertility, decreased libido, impotence
Growth hormone: acromegaly	Enlarged hands and feet, coarse facial features, jaw protrusion, thickened lips, joint pain, sleep apnea, enlarged heart and liver, hyperglycemia . Skeletal changes are permanent
ACTH	Cushing disease (see below)

- Ask about a change in hat, glove, ring, or shoe size, plus headaches and vision changes.
- **Growth hormone suppression test:** oral glucose normally suppresses growth hormone. A level that does not fall below 5 ng/mL is abnormal.

Drugs

- **Bromocriptine, cabergoline** (dopamine agonists): inhibit release of growth hormone and prolactin. Give bromocriptine with a meal or snack. Side effects: orthostatic hypotension, headache, nausea, abdominal cramps, constipation.
- **Octreotide, lanreotide** (somatostatin analogs) and **pegvisomant** (growth hormone receptor blocker) for acromegaly.

Drug alert: bromocriptine

Seek care immediately for **chest pain, dizziness, or watery nasal discharge**. These can signal dysrhythmias, coronary artery spasm, or fibrotic changes.

Hypophysectomy (transsphenoidal or transnasal)

Postoperative nursing management, in order

1. **Neurologic checks hourly for the first 24 hours**, then every 4 hours. Report vision changes, mental status changes, altered level of consciousness, or decreased extremity strength.
2. Keep the **head of the bed elevated**.
3. Inspect the nasal drip pad for the type and amount of drainage (see the cerebrospinal fluid leak box).
4. Monitor fluid balance, especially **output greater than intake** (transient diabetes insipidus).
5. Have the client deep breathe hourly while awake. Instruct: **do not cough, sneeze, blow the nose, or bend at the waist**; these raise intracranial pressure. Bend the knees to pick things up.
6. Mouth care with floss and mouth rinses. **No toothbrushing for about 2 weeks**. Lubricate dry lips; mouth breathing is expected.
7. Prevent constipation and straining: fiber, fluids, stool softeners.
8. Teach self-administration of prescribed hormones. If the whole gland is removed, thyroid hormone and glucocorticoid replacement is **lifelong**.

Expected finding to teach: a decreased sense of smell for 3 to 4 months.

Recognize a cerebrospinal fluid leak

- Client reports **postnasal drip or increased swallowing**.
- **Halo sign**: light yellow ring at the edge of clear drainage on the drip pad.
- Drainage tests positive for **glucose**.
- Persistent severe headache.

Also watch for meningitis: **headache, fever, nuchal rigidity**.

Diabetes insipidus

Newer names used in your book: **arginine vasopressin deficiency** (central diabetes insipidus) and **arginine vasopressin resistance** (nephrogenic diabetes insipidus).

Like you're 10

Antidiuretic hormone is the **drain plug** in the kidney's bathtub. In diabetes insipidus the plug is missing (deficiency) or the drain ignores it (resistance). Water pours out no matter what. The urine looks like water, and what is left in the blood gets too salty.

- **Cues:** polyuria, nocturia, polydipsia. Urine output **more than 3 L in 24 hours**, and it can reach 4 to 30 L per day.
- **Urine:** dilute. Specific gravity **below 1.005**, osmolality below 300.
- **Blood:** plasma osmolality and **sodium increase**. Dehydration, hypotension.
- Urine output does **not** decrease when fluid intake decreases.
- Causes to ask about: recent surgery, head trauma, **lithium** (can cause the resistance type).

Safety priority

Never deprive a client with suspected diabetes insipidus of fluids for more than 4 hours. They cannot reduce urine output, and severe dehydration results.

Desmopressin can cause hyponatremia, leading to seizures, coma, respiratory failure, or death. Monitor serum sodium.

Drugs

- **Desmopressin** replaces vasopressin in the deficiency type. Oral, sublingual, subcutaneous, or intranasal. The intranasal form can cause mucous membrane ulceration and chest tightness; switch routes for an upper respiratory infection.
- **Resistance type:** low-salt, low-to-normal protein diet and a **thiazide diuretic**, which reduces urine output by inducing mild volume depletion.

Client education

- Weigh daily: same scale, same time, similar clothing.
- Weight gain of **more than 2.2 lb (1 kg)** with persistent headache, acute confusion, nausea, or vomiting means water toxicity: go to the emergency department.
- **Polyuria and polydipsia mean another dose is needed.**
- Drink fluids equal to urine output. Wear a medical alert bracelet.

Syndrome of inappropriate antidiuretic hormone

Like you're 10

Now the drain plug is **stuck in**. Water cannot leave. The tub fills, and the salt in it gets watered down, like adding too much water to a drink mix. The sodium is not gone; it is **diluted**. Brain cells swell in the watery blood, which is why the danger is seizures.

- **Causes:** small cell lung cancer and other malignancies; central nervous system trauma, infection, stroke; pneumonia, tuberculosis, positive-pressure ventilation; drugs such as selective serotonin reuptake inhibitors, tricyclics, carbamazepine, cyclophosphamide, vincristine, opiates.
- **Early cues:** loss of appetite, nausea, vomiting; weight gain. **Dependent edema is usually not present**, because it is free water that is retained.

- **Neurologic:** lethargy, headache, hostility, disorientation, decreased deep tendon reflexes, then seizures and coma. Worst when sodium is below 115 mEq/L.
- **Vital signs:** full, bounding pulse; hypothermia.
- **Labs:** serum sodium low (can reach 110). Plasma osmolarity low. Urine volume low; urine osmolarity, urine sodium, and specific gravity **high**.

Nursing management, in order

1. **Assess neurologic status.** Check orientation to time, place, and person every 2 hours; watch for muscle twitching, irritability, and restlessness before they progress to seizures or coma.
2. **Assess for fluid overload at least every 2 hours:** bounding pulse, neck vein distention, crackles, dyspnea. Pulmonary edema can develop quickly; notify the provider of any worsening.
3. **Maintain fluid restriction,** sometimes as low as 500 to 1000 mL per 24 hours.
4. Use **saline, not water,** to dilute tube feedings, irrigate gastrointestinal tubes, and give drugs by tube.
5. **Sodium below 120 mEq/L: provide a safe environment.** Reduce noise and lighting; side rails secured.
6. Measure intake, output, and daily weight. **1 kg gained = 1000 mL retained.** A gain of 2.2 lb or more per day is cause for concern.
7. Offer frequent oral rinsing for dry mouth; warn the client not to swallow the rinse.

Drugs

- **Tolvaptan (oral), conivaptan (IV):** vasopressin receptor antagonists. Promote water excretion without sodium loss. **Give only in the hospital** so sodium can be watched closely. Tolvaptan warnings: too-rapid sodium rise (more than 12 mEq/L in 24 hours) is linked to central nervous system demyelination; liver failure.
- **3% hypertonic saline** when sodium is very low. Give cautiously; it can worsen fluid overload and promote heart failure.
- **Diuretics** only on a limited basis, when sodium is near normal and heart failure is present.

Side by side: the two antidiuretic hormone disorders

	Diabetes insipidus	Syndrome of inappropriate antidiuretic hormone
Antidiuretic hormone	Too little, or kidney does not respond	Too much
Urine volume	Very high	Low
Urine specific gravity	Low (below 1.005)	High
Serum sodium	High	Low (dilutional)
Plasma osmolarity	High	Low
Fluid status	Dehydration, hypotension	Fluid overload, bounding pulse
Fluids	Replace; never withhold more than 4 hours	Restrict
Key drug	Desmopressin	Tolvaptan or conivaptan
Drug danger	Too much causes hyponatremia	Too-rapid sodium rise

Adrenal gland hypofunction (adrenal insufficiency)

Like you're 10

The adrenal glands make two helpers. **Cortisol** keeps sugar in the blood and helps you handle stress. **Aldosterone** holds on to salt and water and throws out potassium. Lose both, and the sugar drops, the salt and water drain away, and potassium piles up.

	Primary (Addison disease)	Secondary
Where is the problem	Adrenal gland itself	Pituitary or hypothalamus
Most common cause	Autoimmune disease	Sudden cessation of long-term glucocorticoid therapy
What is lost	Glucocorticoids, mineralocorticoids, and androgens	Mineralocorticoid function is preserved
ACTH	Elevated	Decreased
Skin	Hyperpigmentation; vitiligo with autoimmune disease	No pigment change
ACTH stimulation test	Cortisol response absent or very decreased	Cortisol response increased

- **Cues:** fatigue, muscle weakness, **salt craving**, anorexia, nausea, vomiting, abdominal pain, weight loss, hypotension.
- **Labs:** sodium **low**, potassium **high**, fasting glucose normal to low, calcium high, blood urea nitrogen high, cortisol low.
- Potassium retention also promotes hydrogen ion reabsorption, which can lead to **acidosis**.
- The ACTH stimulation test is the most definitive test. **If acute adrenal insufficiency is suspected, treatment starts without it.**

Acute adrenal insufficiency (adrenal crisis, Addisonian crisis): emergency

The need for cortisol and aldosterone exceeds supply. Triggered by **surgery, trauma, or severe infection**, especially when output is already low. Sodium falls and potassium rises rapidly; severe hypotension follows.

Emergency management, in order:

1. **Start a rapid infusion** of normal saline or 5% dextrose in normal saline.
2. **Give hydrocortisone or dexamethasone as an IV bolus**, then hydrocortisone by continuous IV infusion over the next 8 hours.
3. **Monitor heart rate, rhythm, and ECG** for hyperkalemia: slow heart rate, heart block, **tall peaked T waves**. Assessing cardiac function is a nursing priority.
4. **Treat hyperkalemia:** IV insulin with dextrose to shift potassium into cells; potassium-binding resin; loop or thiazide diuretics; IV calcium to stabilize the cardiac membrane; potassium restriction. **Avoid potassium-sparing diuretics.**
5. **Treat hypoglycemia:** IV glucose as ordered; glucagon as needed; maintain IV access; check blood glucose hourly.
6. Start an IV H2 blocker (for example, cimetidine) for ulcer prevention.
7. Monitor intake and output.

Drugs

- **Hydrocortisone, cortisone, prednisone** replace cortisol. Divided doses: **two-thirds on arising, one-third at 6 p.m.** (teaching box: second dose between 4 and 6 p.m.), to mimic normal release. Take with meals or a snack.
- **Fludrocortisone** replaces aldosterone. Monitor blood pressure for hypertension; report weight gain or edema.
- **Prednisone and prednisolone are sound-alike and not interchangeable.** Prednisolone is more potent.
- Do not start salt restriction or diuretics without considering that they could trigger an adrenal crisis.

Client education: cortisol replacement

- **Never skip a dose and never stop suddenly.**
- The dose is increased, as the provider directs, for increased physical stress or severe emotional stress. Report illness.
- If persistent vomiting or severe diarrhea prevents oral doses for 24 to 36 hours, call the provider or go to the emergency department; an injection is needed.
- Learn, with a family member, to give IM hydrocortisone.
- Weigh daily. Report rapid weight gain, round face, or fluid retention (too much drug).
- Always wear a medical alert bracelet. Carry simple carbohydrates.

Cushing syndrome (hypercortisolism)

Like you're 10

Now there is **too much** cortisol. Cortisol breaks down muscle, skin, and bone for fuel, parks fat on the belly, face, and upper back, pushes sugar up, holds salt and water, and quiets the immune system. So the body looks big in the middle and thin in the arms and legs, bruises easily, and cannot fight germs.

- **Cushing syndrome** is any cortisol excess. The most common cause is **taking glucocorticoid drugs** such as prednisone.
- **Cushing disease** is the specific type caused by a pituitary tumor secreting ACTH.
- **Appearance:** moon face, buffalo hump, truncal obesity with thin arms and legs, reddish-purple striae, thin skin, bruising, acne, hirsutism.
- **Other cues:** hypertension, edema, muscle wasting, osteoporosis and fractures, poor wound healing, emotional instability, sleep difficulty, oligomenorrhea, gastric ulcers.
- **Labs:** cortisol high in blood, saliva, and urine. Glucose **high**, sodium **high**, lymphocytes **low**, calcium **low**.
- **ACTH level tells you the source:** high in pituitary Cushing disease; low when the cause is an adrenal tumor or steroid drugs.
- **Dexamethasone suppression test:** if cortisol is suppressed, Cushing disease is not present.

Clinical judgment: the three priority problems

1. **Fluid overload** from water and sodium retention.
2. **Potential for injury** from thin skin, poor healing, and bone loss.
3. **Potential for infection** from reduced immunity.

Nursing management, in priority order

1. **Fluid overload (priority problem 1):** monitor at least every 2 hours for bounding pulse, neck vein distention, crackles, edema, reduced urine output. Weigh daily; weight is the best indicator of fluid retention. Maintain sodium and fluid restriction as ordered.
2. **Skin injury (priority problem 2):** assess skin; turn every 2 hours and pad bony prominences; use tape sparingly and remove it carefully; hold pressure after venipuncture; soft toothbrush and electric shaver.
3. **Fractures (priority problem 2):** move the client with a lift sheet; gait belt when walking; high-calorie diet with calcium and vitamin D; avoid caffeine and alcohol.
4. **Gastrointestinal bleeding (priority problem 2):** give H2 blockers (cimetidine, famotidine), proton pump inhibitors (omeprazole), or antacids on a regular schedule. Limit NSAIDs, aspirin, alcohol, caffeine, smoking, and fasting.
5. **Infection (priority problem 3):** strict handwashing and aseptic technique; vital signs at least every 4 hours; assess mouth, lungs, and urine. Cortisol masks the signs, so fever and pus may be absent.

Safety priority

In a client with reduced immunity, **a temperature rise of even 1 F above baseline is significant** and means infection until proven otherwise.

Ketoconazole is teratogenic and can cause liver damage; follow liver function tests.

Drugs that lower cortisol

- **Ketoconazole, levoketoconazole:** inhibit cortisol synthesis.
 - **Cyproheptadine:** interferes with ACTH production.
 - **Mifepristone:** blocks glucocorticoid receptors; for adults with type 2 diabetes who do not respond to other drugs.
 - **Pasireotide:** subcutaneous; for pituitary adenoma. **Osilodrostat:** persistent disease after surgery.
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- **Adrenalectomy:** glucocorticoids are given before and during surgery to prevent adrenal crisis. After surgery, assess every 15 minutes for shock from insufficient glucocorticoid.
 - **Bilateral** adrenalectomy: lifelong glucocorticoid and mineralocorticoid replacement. **Unilateral:** replacement until the remaining gland recovers, up to 2 years.
 - Teach: report weight gain of more than 3 lb in a week or 1 to 2 lb in 24 hours; avoid crowds; yearly influenza vaccine; report fever immediately.

Hyperaldosteronism (primary type: Conn syndrome)

Like you're 10

Too much of the "keep salt, dump potassium" helper. Salt and water stay, so pressure goes up. Potassium and acid get thrown out.

- **Labs:** sodium high, **potassium low**, metabolic **alkalosis**. Renin low, aldosterone high.
- **Cues:** hypertension and hypokalemia are the most common problems; headache, fatigue, muscle weakness; paresthesias if potassium is severely low.
- **Treatment:** adrenalectomy. Potassium must be corrected before surgery.

Drug: spironolactone

Potassium-sparing diuretic and aldosterone antagonist. Risk is now **hyperkalemia**, especially with impaired kidney function.

- Avoid potassium supplements and potassium-rich foods.
- Report hyponatremia symptoms: muscle weakness, dizziness, lethargy, drowsiness.
- Other side effects: gynecomastia, diarrhea, headache, rash, erectile dysfunction, hirsutism, amenorrhea.

The adrenal disorders on one line each

	Sodium	Potassium	Glucose	Blood pressure	Acid-base
Adrenal insufficiency	Low	High	Normal to low	Low	Acidosis
Cushing syndrome	High	Low	High	High	
Hyperaldosteronism	High	Low		High	Alkalosis

Potassium for Cushing syndrome comes from the adrenal hyperfunction column of the chapter's laboratory profile. Blank cells mean the chapter does not make a point of it.

PART 4 | CHAPTER 55: THYROID AND PARATHYROID GLAND CONDITIONS

Priority concept: cellular regulation. Interrelated: nutrition and gas exchange. The thyroid sets the speed of every cell. The parathyroids set the calcium level.

Hypothyroidism

Like you're 10

Thyroid hormone is the **gas pedal** for every cell. In hypothyroidism the foot is off the pedal. The heart beats slower, the gut moves slower, the brain thinks slower, and the body makes less heat. The pituitary notices and yells louder (TSH goes up), which can make the gland swell into a goiter without making more hormone.

- **Most common cause in the United States:** Hashimoto thyroiditis (autoimmune). Also thyroid surgery, radioactive iodine treatment, iodine deficiency, and drugs: **amiodarone**, **lithium**, guaifenesin.
- **Labs:** T3 and T4 **low**. TSH **high** in primary hypothyroidism; low or near normal in secondary.
- **Myxedema:** cellular edema from built-up metabolites. Puffy face, periorbital edema, thick tongue, nonpitting edema.
- A **goiter** suggests a thyroid problem but does not tell you whether it is too much or too little hormone.

Clinical judgment: the three priority problems

1. **Decreased gas exchange** (hypoventilation, muscle weakness).
2. **Reduced perfusion** from decreased cardiac output.
3. **Potential for myxedema coma.**

Respiratory and cardiac problems come first. The most common cause of death in myxedema coma is **respiratory failure**.

Nursing management, in order

1. **Assess breathing:** rate and depth of respirations; oxygen saturation by pulse oximetry; auscultate for decreased breath sounds or crackles.
2. **Apply oxygen** if the client is hypoxemic. Severe cases may need ventilatory support.
3. **Monitor blood pressure, heart rate, and rhythm.** Instruct the client to report chest pain or discomfort immediately.
4. **Avoid sedation** if possible. If it is needed, expect a reduced dose and assess gas exchange afterward.
5. Assess at least every 8 hours for worsening, especially **mental status changes**, and report promptly.

Drug: levothyroxine

- **Lifelong.** Dose is guided by TSH levels and the client's response.
- **Start low and increase slowly**, especially in older adults and clients with coronary disease. Too much too fast can cause dysrhythmias, heart failure, and myocardial infarction.
- Assess for **chest pain and dyspnea** when therapy starts.
- Take with water on an **empty stomach, 30 to 60 minutes before breakfast**.
- Separate from **iron, calcium supplements, and antacids by at least 4 hours**. Fiber supplements can also interfere with absorption.
- Do not change the dose or schedule, and do not take over-the-counter drugs, without asking the provider.

Client education

- Know the symptoms of **both** hypothyroidism and hyperthyroidism; too much replacement causes the second.
- Self-monitor with two easy signs: **sleep and bowels**. More sleep and constipation: the dose may be too low. Trouble sleeping and more stools: the dose may be too high.
- Fiber and fluids for constipation; extra clothing for cold intolerance; medical alert bracelet; periodic blood tests.

Myxedema coma: emergency

Triggers: acute illness, surgery, chemotherapy, stopping thyroid replacement, sedatives or opioids.

Findings: greatly reduced level of consciousness, respiratory failure, hypotension, hyponatremia, hypothermia, hypoglycemia, bradycardia.

Treat on symptoms; **do not wait for laboratory confirmation**.

Emergency management, in order:

1. **Maintain a patent airway.**
2. Replace fluids with IV normal or hypertonic saline as ordered.
3. Give **IV T3 and/or T4** as prescribed.
4. Give IV glucose as prescribed.
5. Give corticosteroids as prescribed.
6. Check temperature and blood pressure hourly. Cover with warm blankets.
7. Monitor for changes in mental status.
8. Turn every 2 hours. Institute aspiration precautions.

Hyperthyroidism

Like you're 10

Now the gas pedal is **stuck down**. The heart races, the body overheats, food burns off before it can be stored, and the nerves are jumpy. In Graves disease, the immune system makes a fake "go" signal that sits on the thyroid's TSH receptor and never lets up, so the real TSH drops to low.

- **Most common cause:** Graves disease (autoimmune). Also toxic multinodular goiter and too much thyroid replacement hormone.
- **Graves disease only:** exophthalmos and pretibial myxedema.
- **Labs:** T3 and T4 **high**; TSH **low** in Graves disease.
- **Heat intolerance is often the first symptom.** Also weight loss with increased appetite, more stools, palpitations, tachycardia, atrial fibrillation, higher systolic pressure with a widened pulse pressure, tremors, insomnia, fine silky hair, warm moist skin, hyperglycemia.
- **Eye findings in all types:** eyelid lag and globe lag.

Body system	Hypothyroidism (slow, cold)	Hyperthyroidism (fast, hot)
Heart	Bradycardia, diastolic hypertension, decreased activity tolerance	Tachycardia, dysrhythmias, palpitations, increased systolic pressure
Temperature	Cold intolerance, low body temperature	Heat intolerance, diaphoresis

Body system	Hypothyroidism (slow, cold)	Hyperthyroidism (fast, hot)
Weight and gut	Weight gain, constipation	Weight loss with increased appetite, increased stools
Skin and hair	Cool, pale, dry, coarse; brittle hair; loss of eyebrow hair	Warm, moist, smooth; fine soft hair
Nerves and mind	Lethargy, slow speech, impaired memory, decreased reflexes, depression	Tremors, hyperactive reflexes, insomnia, anxiety, restlessness
TSH (primary or Graves)	High	Low
Crisis	Myxedema coma	Thyroid storm

Nursing management, in order

1. **Monitor** apical pulse, blood pressure, and temperature at least every 4 hours. **Report a temperature rise of even 1 F immediately**, then assess cardiac status.
2. Instruct the client to report palpitations, dyspnea, vertigo, or chest pain immediately.
3. **Reduce stimulation:** encourage rest; quiet room, closed door, limited visitors; postpone nonessential care.
4. **Promote comfort:** cool room, fresh ice water, change damp linen, cool showers or sponge baths.
5. **Protect the eyes:** artificial tears; elevate the head of the bed at night; dark glasses for photophobia; gently tape lids closed at bedtime if they do not close.

Drugs

- **Methimazole (first line), propylthiouracil:** block thyroid hormone production. Response is delayed because stored hormone keeps releasing. They reduce the immune response: avoid crowds and ill people. Watch for hypothyroid signs (weight gain, slow heart rate, cold intolerance).
- **Methimazole causes birth defects;** notify the provider if pregnancy occurs.
- **Propylthiouracil is liver-toxic;** report dark urine or yellow skin or eyes. The two drugs are not interchangeable.
- **Iodine (saturated solution of potassium iodide):** short-term before surgery; decreases blood flow through the gland. Give diluted in a beverage, **1 hour after** the thionamide.
- **Propranolol, atenolol:** relieve diaphoresis, anxiety, tachycardia, palpitations. They do not lower hormone production.

Client education: radioactive iodine therapy

Outpatient. Relief may take 6 to 8 weeks. **Not used in pregnancy.** Hypothyroidism can follow, even years later, so follow-up is lifelong.

- Sit to urinate; flush two to three times with the lid closed.
- Stay at least 6 feet from pregnant women, infants, and young children for the prescribed time; do not share a bed.
- Do not share toothbrushes, toothpaste, food, or drinks; do not prepare food for others.

Thyroidectomy

Thyroidectomy: nursing management, in order

Before surgery: the client is made euthyroid with a thionamide; iodine reduces gland size and vascularity. Teach the client to **support the neck with both hands** behind it when coughing or moving.

After surgery:

1. **Vital signs every 15 minutes** until stable, then every 30 minutes.
2. **Position semi-Fowler** while awake, pillows supporting head and neck. **Avoid neck extension.**
3. **Keep tracheostomy equipment, oxygen, and suction at the bedside** and confirm they work.
4. Assess breathing: listen for **stridor**; watch for dyspnea and falling oxygen saturation. Help the client deep breathe every 30 minutes to 1 hour; suction as needed.
5. Inspect the dressing **and behind the neck** for blood.
6. Ask about **tingling around the mouth, toes, and fingers**; watch for muscle twitching.
7. Assess the voice every 2 hours for hoarseness.
8. Monitor temperature; report a rise of even 1 F.
9. Assess pain and give prescribed analgesics.

Complication	What you find	What you do
Hemorrhage (first 24 hours)	Blood on the dressing or behind the neck ; respiratory distress from tracheal compression	Inspect front and back of the neck. A moderate amount of serosanguineous drain output is normal
Respiratory distress	Stridor , dyspnea, falling oxygen saturation, inability to swallow, drooling	Initiate the Rapid Response Team. Keep tracheostomy equipment, oxygen, and suction at the bedside
Hypocalcemia and tetany (parathyroid injury)	Tingling around the mouth, toes, and fingers ; muscle twitching	Have IV calcium gluconate available
Laryngeal nerve damage	Hoarseness, weak voice	Assess the voice every 2 hours. Hoarseness is usually temporary
Thyroid storm	Fever, tachycardia	Report a temperature rise of even 1 F immediately

Thyroid storm: emergency

Uncontrolled hyperthyroidism, most often Graves disease. **Key findings: high fever and tachycardia.** Also abdominal pain, nausea, vomiting, diarrhea, anxiety, tremors, then restlessness, confusion, seizures, coma.

Triggers: trauma, infection, diabetic ketoacidosis, pregnancy, vigorous palpation of the goiter, iodine exposure, radioactive iodine therapy.

Emergency management, in order:

1. **Maintain a patent airway and adequate ventilation.**
2. Give **antithyroid drugs** as prescribed: methimazole or propylthiouracil.
3. Give iodide solution and glucocorticoids as prescribed.
4. Give **propranolol** as prescribed to control heart rate.
5. **Monitor continually for dysrhythmias.** Vital signs every 15 to 30 minutes.
6. Reduce fever: **nonsalicylate** antipyretic; cooling blanket or ice packs.
7. Correct dehydration with normal saline infusions as ordered.

- **Thyroiditis:** Hashimoto thyroiditis is the chronic autoimmune form. Painless enlargement, dysphagia, elevated antithyroid peroxidase antibody. Subacute thyroiditis follows a viral infection and typically runs hyperthyroid, then hypothyroid, then normal.
- **Thyroid cancer:** first sign is usually a single painless nodule. Papillary is the most common and slow growing; anaplastic is rare and aggressive. Surgery is the treatment of choice. The client is hypothyroid afterward and needs replacement teaching.

Hypoparathyroidism

Like you're 10

Parathyroid hormone is the worker who keeps the calcium shelf stocked. With no worker, calcium runs low. Nerves and muscles need calcium to stay calm. Without it they fire on their own: tingling first, then cramps and spasms (tetany).

- **Most common cause:** after thyroid or parathyroid removal, or head and neck cancer surgery.
- **Cues, mild to severe:** tingling and numbness around the mouth and in hands and feet; severe muscle cramps; spasms of hands and feet; seizures. Positive **Trousseau and Chvostek signs** point to tetany.
- **Labs:** calcium low, **phosphorus high**, parathyroid hormone low, magnesium low, vitamin D low.

Drugs

- **IV calcium gluconate** for acute, severe hypocalcemia.
- **Oral calcitriol and calcium carbonate;** vitamin D for milder cases. Long-term daily calcium.
- The dose keeps calcium in the **low-normal** range: high enough to stop symptoms, low enough to avoid high urine calcium and **kidney stones**.

Client education

Therapy is lifelong. Eat high-calcium foods (milk, yogurt, cheese) in moderation. Wear a medical alert bracelet.

Hyperparathyroidism

Like you're 10

Now the worker will not stop. To keep stocking the blood shelf, the worker **pulls calcium out of the bones**. Bones get thin. The extra calcium in the blood clogs the kidneys as stones and slows the gut and the brain.

- **Labs:** calcium **high**, phosphorus **low**, parathyroid hormone high.
- **Cues:** kidney stones, bone pain and fractures, anorexia, nausea, vomiting, epigastric pain, constipation, peptic ulcer disease (hypercalcemia raises gastrin), fatigue, lethargy. Calcium above 12 mg/dL: psychosis, confusion, then coma.
- **Secondary hyperparathyroidism:** common causes are **vitamin D deficiency and chronic kidney disease**.
- **Treatment of choice:** parathyroidectomy. Nursing care is the same as for thyroidectomy.

Drugs

- **Cinacalcet** (calcimimetic): reduces parathyroid hormone production and lowers serum calcium. Calcium is monitored routinely.
- **Bisphosphonates** for clients with osteoporosis or fracture risk who are not having surgery.

Client education and safety

- Regular exercise; adequate hydration; avoid prolonged bed rest; moderate calcium and vitamin D intake.
- Avoid lithium if possible, because it can raise calcium.
- Bones are fragile: handle gently and reposition with a **lift sheet**.

Connection: the parathyroid and the kidney

In chronic kidney disease the kidney cannot excrete phosphorus and cannot activate vitamin D. Phosphorus rises, calcium falls, and the parathyroid glands respond by releasing more hormone, which pulls calcium from bone. That is **renal osteodystrophy**, and it is why cinacalcet appears in both Chapter 55 and Chapter 60.

PART 5 | CHAPTER 56: DIABETES MELLITUS

Priority concept: glucose regulation.

Like you're 10

Sugar is the fuel. Insulin is the **key that opens the cell door** so fuel can get in. In **type 1**, the key factory (beta cells) was destroyed, so there are no keys. In **type 2**, there are keys, but the locks are rusty (insulin resistance) and the factory is wearing out.

Either way, sugar is stuck in the hallway (blood) while the cells starve. The kidneys try to dump the extra sugar in the urine, and sugar drags water out with it. That is why the classic signs are lots of urine, lots of thirst, and lots of hunger.

Feature	Type 1	Type 2
Pathology	Autoimmune beta cell destruction; absolute insulin deficiency	Insulin resistance with dysfunctional beta cells
Age at onset	Usually younger than 30	Any age in adults
Onset and symptoms	Abrupt: thirst, hunger, increased urine output, weight loss	Frequently none; thirst, fatigue, blurred vision, vascular or nerve complications
Ketosis	Ketosis-prone	Ketosis-resistant
Body weight	Same obesity prevalence as general population	60% to 80% obese
Insulin	All dependent on insulin	Required for 20% to 30%
Acute crisis	Diabetic ketoacidosis	Hyperglycemic-hyperosmolar state

Diagnosis and monitoring

Test	Normal	Prediabetes	Diabetes
Glycosylated hemoglobin (A1C)	4% to 5.7%	5.7% to 6.4%	6.5% or higher
Fasting blood glucose (no calories for at least 8 hours)	74 to 100 mg/dL	100 to 125 mg/dL	126 mg/dL or higher
2-hour glucose tolerance	Below 140 mg/dL	140 to 199 mg/dL	200 mg/dL or higher
Random glucose with classic symptoms or crisis			Above 200 mg/dL

- Without unequivocal hyperglycemia, diagnosis needs **two abnormal results**.
- **A1C target: below 7%** without significant hypoglycemia, for nonpregnant adults. Check at least twice a year if stable, quarterly if therapy changed or not at target.
- **Continuous glucose monitoring:** time-in-range is 70 to 180 mg/dL; goal is 70% of the time.
- Target blood glucose after treating hypoglycemia: 70 to 110 mg/dL.

Non-insulin drugs

Class and examples	How it works	Nursing implications
Biguanide: metformin (first line for type 2)	Lowers liver glucose production; improves insulin sensitivity	No alcohol (lactic acidosis). Stop before imaging with contrast. Take with food for gastrointestinal upset
GLP-1 agonists: dulaglutide, exenatide, liraglutide, semaglutide	Reduce glucagon and liver glucose production; delay gastric emptying	Mostly subcutaneous; some are weekly. Report persistent abdominal pain and nausea (pancreatitis). Contraindicated in gastroparesis
SGLT2 inhibitors: canagliflozin, dapagliflozin, empagliflozin	Kidney excretes filtered glucose and sodium in the urine	Teach signs of dehydration, hyponatremia, urinary tract infection, genital yeast infection. Report genital or perineal swelling, tenderness, redness (Fournier gangrene)
DPP-4 inhibitors: sitagliptin, saxagliptin, linagliptin, alogliptin	Keep natural gut hormones working longer	Watch for rash or allergy, pancreatitis, and heart failure symptoms
Sulfonylureas: glipizide, glyburide, glimepiride. Meglitinides: repaglinide, nateglinide	Trigger insulin release from beta cells	Cause hypoglycemia. Take with or just before meals. Many drug interactions
Thiazolidinediones: pioglitazone, rosiglitazone	Improve insulin receptor sensitivity	Heart failure risk: daily weight; report gain of more than 2 lb in a day or 4 lb in a week. Report vision changes (macular edema)
Alpha-glucosidase inhibitors: acarbose, miglitol	Delay carbohydrate absorption in the intestine	Take only with a meal. Bloating, flatulence, diarrhea
Amylin analog: pramlintide	Delays gastric emptying; triggers satiety	Never mix with insulin in the same syringe

Insulin

Type	Examples	Onset	Peak	Duration
Rapid-acting	Aspart, glulisine, lispro	10 to 30 minutes	0.5 to 3 hours	3 to 6 hours
Short-acting	Regular	30 to 60 minutes	1 to 5 hours	6 to 10 hours
Intermediate-acting	NPH	60 to 120 minutes	6 to 14 hours	16 to 24 hours
Long-acting	Glargine, detemir	60 to 120 minutes (glargine about 70)	No peak	12 to 24 hours
Ultra-long-acting	Degludec	30 to 90 minutes	No peak	24 hours or more

- **Basal insulin** (intermediate, long, or ultra-long) covers between meals and makes up about 50% of the daily dose. **Prandial insulin** (rapid or regular) covers meals.
- Rapid-acting insulin is given **within 10 minutes before the meal.**
- NPH plus regular twice daily: two-thirds of the dose before breakfast, one-third before the evening meal. Because NPH peaks, a **bedtime snack** may be needed to prevent nighttime hypoglycemia.
- Inject subcutaneously at **90 degrees**; 45 degrees for frail older adults and very thin clients. **No aspiration.**

- NPH is uniformly **cloudy** after gently rolling the vial. Other insulins should be clear. Inspect for clumping, frosting, or color change.
- Refrigerate insulin not in use. In-use insulin may stay at room temperature up to 28 days; discard after 28 days. Never freeze.
- Do not exercise within 1 hour of an injection or near the insulin's peak.

Pearl: two causes of morning hyperglycemia

- **Dawn phenomenon:** nighttime hormone release raises glucose around 5 to 6 a.m. Fix: **more insulin overnight** (for example, move the evening intermediate-acting dose to 10 p.m.).
- **Somogyi phenomenon:** rebound from **nighttime hypoglycemia**. Fix: adequate **bedtime snack** and review of the insulin dose and exercise.

Both are diagnosed by checking blood glucose during the night.

Hypoglycemia

Like you're 10

The brain runs on sugar and cannot store it. When sugar drops, the body first sounds an alarm (shaky, sweaty, hungry, fast heartbeat). If nobody answers the alarm, the brain itself starts shutting down: confusion, seizure, coma.

Level	Findings	Treatment
Mild (usually below 70 mg/dL)	Hungry, irritable, shaky, weak, headache; fully conscious	15 g carbohydrate. Recheck in 15 minutes. Repeat if still below 70. Then a small complex carbohydrate and protein snack
Moderate (usually below 50 mg/dL)	Cold clammy skin, pallor, rapid pulse, mood change, drowsiness	30 g rapidly absorbed carbohydrate. Glucagon may be indicated. Recheck in 15 minutes
Severe	Unable to swallow, confused, combative, unconscious, seizures	Glucagon IM, subcutaneous, or nasal; repeat in 10 minutes if still unconscious. In hospital: concentrated IV dextrose. Notify the provider

Hypoglycemia: nursing management, in order

Client is conscious and can swallow (the 15-15 rule):

1. Give **15 g of carbohydrate** (30 g if glucose is below 50 mg/dL). A liquid form is preferred.
2. **Recheck blood glucose in 15 minutes.**
3. If still low, **repeat the same treatment.**
4. Once above the hypoglycemic range, give a **complex carbohydrate** such as crackers with nut butter.

Severe hypoglycemia (cannot swallow): nursing management, in order

1. Give **concentrated IV dextrose** (hospital) or **glucagon** IM, subcutaneous, or nasal. Position the client **on the side**; glucagon can cause vomiting.
2. **Repeat glucagon in 10 minutes** if the client remains unconscious.
3. **Notify the diabetes health care provider** immediately and follow instructions.
4. Check blood glucose every 15 minutes until hypoglycemia resolves, then monitor for several hours.
5. Give a small meal when the client is awake and no longer nauseated.

- 15 g examples: glucose tablets or gel, half a cup of fruit juice or regular soft drink, 5 hard candies, 4 teaspoons of sugar, 1 tablespoon of honey.
- Give concentrated dextrose carefully; it damages tissue if it extravasates. In the hospital, avoid high-potassium options such as orange juice.
- **Causes:** too much insulin, missed or delayed meals, exercise, **alcohol**, gastroparesis, and **decreased insulin clearance from progressive kidney failure**.
- Alcohol only with or shortly after a meal. Delaying a meal more than 30 minutes raises the risk with some regimens.

	Hypoglycemia	Hyperglycemia
Skin	Cool, clammy, sweaty	Warm, dry
Dehydration	Absent	Present
Respirations	No consistent change	Rapid, deep (Kussmaul), fruity odor
Mental status	Anxious, nervous, irritable, confused, seizures, coma	Alert to stuporous to coma
Ketones	Negative	Positive

Diabetic ketoacidosis

Like you're 10

With **no insulin at all**, cells cannot get sugar, so the body burns fat instead. Burning fat fast leaves behind acids called ketones. The blood turns acidic. The lungs try to help by breathing deep and fast to blow off acid (Kussmaul breathing), and the breath smells fruity. Meanwhile sugar pours out in the urine and drags water with it.

Hyperglycemic-hyperosmolar state

Like you're 10

Here there is **a little insulin**: enough to stop the fat-burning and the acid, but not enough to control sugar. So sugar climbs far higher, the kidneys dump water for days, and the blood becomes thick like syrup. Thick blood pulls water out of brain cells. The problem is not acid. It is **dehydration and the brain**.

	Diabetic ketoacidosis	Hyperglycemic-hyperosmolar state
Onset	Sudden	Gradual

	Diabetic ketoacidosis	Hyperglycemic-hyperosmolar state
Typical client	Type 1	Older adult with type 2, often undiagnosed
Precipitating factors	Infection, other stressors, inadequate insulin dose	Infection, other stressors, poor fluid intake
Serum glucose	Above 300 mg/dL	Above 600 mg/dL
Osmolarity	Variable	Above 320 mOsm/L
Ketones (serum and urine)	Positive	Negative
Serum pH	Below 7.35	Above 7.4
Bicarbonate	Below 15 mEq/L	Above 20 mEq/L
Hallmark symptoms	Kussmaul respirations, fruity breath, nausea, abdominal pain	Altered central nervous system function: confusion to coma, seizures, reversible paralysis
Both	Polyuria, polydipsia, weight loss, dry skin, lethargy; blood urea nitrogen and creatinine elevated from dehydration	Same

Clinical judgment: diabetic ketoacidosis, in order

First assess airway, level of consciousness, hydration status, electrolytes, and blood glucose.

1. **Fluids first.** Restore volume: 15 to 20 mL/kg/hr in the first hour. Then hypotonic fluids at 4 to 14 mL/kg/hr.
2. **When glucose reaches 250 mg/dL, switch to 5% dextrose in 0.45% saline** to prevent hypoglycemia and cerebral edema.
3. **Regular insulin by continuous IV infusion** after an IV bolus. Goal: glucose falls about 50 to 75 mg/dL per hour.
4. **Potassium.** Mild to moderate hyperkalemia is common at first. Insulin, fluids, and correction of acidosis **drive potassium down**. Replacement starts once the level falls below 5.0 mEq/L.
5. **Bicarbonate only for severe acidosis:** pH 7.0 or less, or bicarbonate below 5 mEq/L.
6. Subcutaneous insulin starts when the client can take oral fluids and ketosis has stopped.

Resolved when: glucose below 200, bicarbonate above 18, venous pH above 7.30, anion gap below 12.

Safety priority

Before any IV potassium, confirm urine output of at least 30 mL/hr.

Watch for hypokalemia during treatment: fatigue, muscle weakness, shallow respirations, abdominal distention or ileus, weak pulse, ECG changes.

Vital signs every 15 minutes until stable. Watch for fluid overload: the first 24 hours may require 6 to 10 L.

Clinical judgment: hyperglycemic-hyperosmolar state, in order

1. **First priority: increase blood volume.** Normal saline in shock or severe hypotension; otherwise half-normal saline. Infuse at 1 L/hr until central venous pressure rises or blood pressure and urine output are adequate.
2. **Reduce the rate to 100 to 200 mL/hr.** Half the fluid deficit is replaced in the first 12 hours, the rest over the next 36 hours.
3. **Assess hourly for cerebral edema:** abrupt mental status change, abnormal neurologic signs, coma.
4. **Notify the provider immediately** for a change in level of consciousness, a change in pupil size, shape, or reaction, or a seizure.
5. Monitor body weight, urine output, kidney function, and for pulmonary congestion and jugular venous distention; these set the infusion rate.

How to read the response: no improvement in consciousness may mean fluids are too slow. **Getting worse after getting better** may mean osmolarity is dropping too fast. Goal: rehydrate and normalize glucose within 36 to 72 hours.

Coma occurs once osmolarity exceeds 350 mOsm/L; normal is 270 to 300. Drugs that can contribute: glucocorticoids, diuretics, phenytoin, beta blockers, calcium channel blockers.

Client education: sick-day rules

- Notify the provider that you are ill.
- Check blood glucose at least **every 2 to 4 hours**.
- Test urine for **ketones**, even if glucose is in range, especially when vomiting.
- **Keep taking insulin or other antidiabetic drugs** unless told otherwise. Insulin is not omitted during illness.
- Drink 8 to 12 ounces of sugar-free fluid every hour while awake. If glucose is below target, drink fluids with sugar.
- Eat at regular times; if solids are not tolerated, take liquids equal to the usual carbohydrate.
- **Call for:** persistent nausea and vomiting, persistent hypoglycemia, moderate or high ketones, glucose still elevated after two supplemental insulin doses, temperature of 101.5 F or fever longer than 24 hours.

Chronic complications

- **Diabetic nephropathy:** diabetes is the leading cause of end-stage kidney disease. Earliest sign is persistent **albuminuria** (30 to 299 mg per 24 hours). Screen urine albumin yearly: from diagnosis in type 2, after 5 years in type 1.
- **Neuropathy:** burning, tingling, then loss of sensation, starting in the toes. Autonomic forms: gastroparesis, neurogenic bladder, orthostatic hypotension, erectile dysfunction, and **loss of the warning signs of hypoglycemia**.
- **Feet:** neuropathy is the main risk factor for an ulcer; poor perfusion is the main cause of poor healing.

Protecting the kidney

- **ACE inhibitors or angiotensin receptor blockers** for persistent albuminuria. Monitor potassium for **hyperkalemia**. Not recommended when blood pressure and albumin excretion are normal.
- Avoid nephrotoxic drugs: aminoglycoside antibiotics, antifungals, **NSAIDs** such as ibuprofen and naproxen.
- Hydrate before and after **contrast media**.
- When dialysis starts, the **insulin dose is reduced**; hypoglycemia risk rises.

Client education: foot care and exercise

- Inspect feet daily, including between the toes. Wash with lukewarm water; dry thoroughly.
- Moisturizer on the feet, **not between the toes**. Clean cotton socks daily.
- Check shoes for objects before wearing; break in new shoes gradually; buy shoes later in the day.
- Trim nails **straight across**. Check bath water with the wrist or a thermometer.
- Never go barefoot. No heating pads or hot-water bottles. No open-toe sandals. Do not self-treat corns or calluses. Do not cross the legs.
- **Exercise (type 1, vigorous):** only when glucose is 100 to 250 mg/dL and urine ketones are absent. Carry a simple sugar and diabetes identification.

PART 6 | CHAPTER 60: ACUTE KIDNEY INJURY AND CHRONIC KIDNEY DISEASE

Priority concept: elimination. Interrelated: acid-base balance, fluid and electrolyte balance, immunity, perfusion.

Like you're 10

The kidneys are the body's **filter and trash service**. Blood flows in, the filter keeps the good stuff, and the trash and extra water leave as urine. The kidneys also balance salt, potassium, and acid, help make red blood cells, and switch on vitamin D.

Acute kidney injury is the trash service stopping suddenly, over hours or days. It can come back. **Chronic kidney disease** is the service slowly shutting down over months to years. It does not come back.

	Acute kidney injury	Chronic kidney disease
Onset	Sudden: hours to days	Gradual: months to years
Course	May not progress; full recovery possible	Progressive and permanent; longer than 3 months
Nephrons involved	50% to 95%	Symptoms at about 75% loss; dialysis at 90% to 95%
Outcome	Good when function returns; 10% to 20% reach end-stage kidney disease	End-stage kidney disease is fatal without dialysis or transplant

Acute kidney injury

- **Definition (any one):** serum creatinine rises 0.3 mg/dL or more within 48 hours; or creatinine reaches 1.5 times baseline within 7 days; or urine output is below 0.5 mL/kg/hr for 6 hours.
- **Oliguria:** urine output below 400 mL/day. **Azotemia:** buildup of nitrogenous wastes in the blood.
- Common: about 20% of hospitalized clients and 60% of intensive care clients. Higher risk: older adults and those with diabetes, hypertension, liver disease, or existing chronic kidney disease.

Like you're 10: the three types

Picture a kitchen sink with a filter. Three things can stop it from draining.

- **Prerenal:** not enough water is coming **to** the filter. The tap is turned down.
- **Intrarenal (intrinsic):** the **filter itself** is damaged.
- **Postrenal:** the **drain pipe after** the filter is blocked, so everything backs up.

Type	What is wrong	Causes
Prerenal	Reduced perfusion to the kidney	Shock, dehydration, blood or fluid loss, burns, sepsis, heart disease, blood pressure medications, NSAIDs, liver failure, anaphylaxis, renal artery stenosis
Intrarenal	Damage to kidney tissue	Nephrotoxic agents (certain antibiotics, chemotherapy, contrast media), glomerulonephritis, pyelonephritis, lupus, clots in renal vessels
Postrenal	Urine flow obstruction	Kidney stones, enlarged prostate, tumors (bladder, prostate, cervical, colon), strictures, nerve damage to the bladder

Connection: the kidney uses endocrine tools to save itself

When perfusion drops, the kidney constricts its vessels, turns on the **renin-angiotensin-aldosterone** pathway, and triggers **antidiuretic hormone** release. These are the Chapter 53 and 54 hormones. They raise blood volume, but they also shut down urine output. That is why low perfusion produces oliguria and azotemia.

Safety priority: prevention is the nursing job. Steps in order

1. **Assess for volume depletion**, a common cause: low urine output, decreased systolic pressure, decreased pulse pressure, orthostatic hypotension, thirst, rising osmolarity.
2. **Intervene early**: give oral fluids, or request an increase in the IV rate if the client cannot take them.
3. **Report urine output below 30 mL/hr for 2 hours, or dark amber urine**. Do not wait 6 hours for the formal definition.
4. **Report a significant creatinine rise urgently**, especially over hours to days.
5. **Question any order for a potentially nephrotoxic drug** and validate the dose. Two or more together raise the risk dramatically. Monitor creatinine, blood urea nitrogen, and drug peak and trough levels.

Contrast media: nursing management, in order

1. **Before**: confirm kidney function has been assessed and that the radiologist and provider know about any reduced function, so the lowest dose is used.
2. **Hold metformin** and other nephrotoxic drugs at least 24 hours before the procedure, if possible.
3. Give prescribed **IV fluids before** the procedure.
4. **After**: continue IV fluids to dilute and flush the contrast.
5. Monitor urine output. Desired outcome: 150 mL/hr for the first 6 hours.
6. Keep metformin on hold for at least 24 hours after the procedure.

- **Complications**: metabolic acidosis, **hyperkalemia**, hyponatremia, hypocalcemia, pulmonary and peripheral edema, heart failure, hypertension, pericarditis, seizures, mental status changes, bleeding, anemia, infection.
- **Not every client is oliguric**. Some have high-volume output. In this diuretic phase the problems reverse: **hypovolemia and electrolyte loss**, so care shifts to replacement. The onset of polyuria can signal the start of recovery.
- Fluid plan depends on the phase: **restrict** when fluid overload is present; give **liberally** to prevent injury or flush contrast.
- **Nutrition**: kidney-specific formulas are lower in sodium, potassium, and phosphorus and higher in calories.

When kidney replacement therapy is indicated

- Symptomatic uremia: pericarditis, neuropathy, decline in cognition.
- Persistent or rapidly rising potassium **above 6.5 mEq/L**.
- Severe metabolic acidosis: **pH below 7.1**.
- Fluid overload that inhibits tissue perfusion.

The temporary dialysis catheter is **not** used for blood samples, drugs, fluids, or central venous pressure monitoring.

Chronic kidney disease

Stage	Estimated glomerular filtration rate	Focus
1: at risk; normal function but abnormal urine findings	Above 90	Screen and manage risk: uncontrolled hypertension, poorly controlled diabetes, nephrotoxic exposure
2: slightly reduced	60 to 89	Reduce risk factors. Dilute urine output can increase and cause dehydration
3: moderately reduced	30 to 59	Slow progression. Restrict fluids, protein, electrolytes
4: severely reduced	15 to 29	Manage complications; prepare for kidney replacement therapy
5: end-stage kidney disease	Below 15	Kidney replacement therapy or transplant

What breaks, and why

Function lost	Result	Plain version
Waste removal	Azotemia, then uremia : metallic taste, anorexia, nausea, vomiting, muscle cramps, hiccups, fatigue, paresthesias, uremic frost, edema, dyspnea	Trash piles up and makes the whole body feel sick
Sodium and water	Early: sodium is lost (hyponatremia risk, polyuria). Late: sodium and water are retained, causing hypertension and edema	First the filter leaks salt; later it cannot let salt go
Potassium excretion	Holds until urine output falls below 500 mL per 24 hours, then climbs fast to 7 or 8 mEq/L. Life-threatening dysrhythmias; tall peaked T waves	The kidney is the main exit for potassium
Acid excretion	Metabolic acidosis ; low bicarbonate; Kussmaul respirations as the lungs compensate	Acid has no exit, so the lungs blow it off
Phosphorus excretion and vitamin D activation	Phosphorus high, calcium low, parathyroid hormone high, calcium pulled from bone: renal osteodystrophy . Itching and calcium-phosphorus deposits	Bones are robbed to fix the blood
Erythropoietin	Anemia	No signal to make red blood cells
Drug clearance	Drugs last longer. Insulin needs drop ; opioid effects last longer	The trash service also removes medicine

Drugs

- **Loop diuretics (furosemide, bumetanide):** only while urine is still made; not effective in end-stage disease. Expected output exceeds intake by 500 to 1000 mL per 24 hours. Potassium loss can be a desired effect.
- **Phosphate binders (calcium acetate, calcium carbonate, sevelamer, lanthanum):** take **with meals**. Not within 2 hours of other drugs. Monitor phosphorus and calcium; cause **constipation**. Report muscle weakness, slow or irregular pulse, confusion (hypophosphatemia).
- **Vitamin D (calcitriol, paricalcitol):** monitor serum calcium.
- **Erythropoietin-stimulating agents (epoetin alfa, darbepoetin alfa):** monitor hemoglobin; dosed to a level no higher than 10 to 11 g/dL. Too high thickens the blood and causes hypertension and risk for myocardial infarction. Report chest pain, difficulty breathing, high blood pressure, rapid weight gain, seizures, swelling.
- **Iron:** oral with meals plus a stool softener; stool color changes. IV iron dextran needs a test dose.
- **Cinacalcet:** lowers parathyroid hormone; monitor calcium and phosphorus.
- **Vitamins and folic acid:** take **after dialysis** so they are not removed.

Safety priority: drugs in kidney disease

- **No magnesium-containing antacids.** The kidneys cannot excrete magnesium.
- **Opioids:** give cautiously; effects last longer and uremic clients are sensitive to respiratory depression.
- **Insulin and antidiabetic drugs:** doses often need reduction; hypoglycemia risk rises.
- Antibiotics that are kidney-excreted or nephrotoxic need dose adjustment. Prophylactic antibiotics are given before dental procedures.
- No salt substitutes made of potassium chloride.

Diet

	Chronic uremia, no dialysis	Hemodialysis	Peritoneal dialysis
Protein	0.55 to 0.60 g/kg/day	1.0 to 1.5 g/kg/day	1.2 to 1.5 g/kg/day
Fluid	Depends on urine output	500 to 700 mL/day plus urine output	Based on weight gain and blood pressure
Potassium	60 to 70 mEq/day	70 mEq/day	Usually no restriction
Sodium	1 to 3 g/day	2 to 4 g/day	Based on weight gain and blood pressure
Phosphorus	700 mg/day	700 mg/day	800 mg/day

Pearl: protein goes UP once dialysis starts

Before dialysis, protein is restricted to limit waste. On dialysis, protein is lost in the process, so the allowance increases. It is highest on peritoneal dialysis because protein is lost in the dialysate. Low prealbumin means poor protein intake.

Hemodialysis

Like you're 10

A machine does the filtering outside the body. Blood leaves through one needle, runs past a filter with cleaning fluid on the other side, and returns through a second needle. To do that several times a week, the surgeon joins an artery to a vein in the arm (a fistula) so the vein grows big and strong enough.

Arteriovenous fistula or graft: assessment, in order

1. **Check circulation:** distal pulses and capillary refill in the arm with the access.
2. **Palpate for a thrill and auscultate for a bruit** over the access, every 4 hours while awake. The client checks daily.
3. Check needle sites for bleeding and signs of infection.

Protect the access at all times: no blood pressure, venipuncture, or IV line in that arm, and never use it for general IV fluids or drugs. Elevate the arm after surgery; encourage range-of-motion exercises. Teach the client not to carry heavy objects with that arm, wear anything that compresses it, or sleep on it.

A new fistula takes 8 to 12 weeks to mature before it can be used.

Hemodialysis: nursing management, in order

Before the session:

1. Hold dialyzable drugs until **after** the session. Ask the nephrology team whether to hold antihypertensives; short-acting ones can add to hypotension during dialysis.
2. Record baseline vital signs and weight.

After the session:

3. **Obtain vital signs and weight** and compare with pre-dialysis values. Blood pressure and weight should be lower.
4. **Monitor for hypotension**, headache, nausea, vomiting, dizziness, and muscle cramps. Hypotension may need IV normal saline.
5. **Monitor for bleeding** for at least 1 hour. Heparin raises the risk: **no invasive procedures for 4 to 6 hours**.
6. Check temperature. A slight elevation is from blood warming; a true fever suggests sepsis and needs blood cultures.
7. Give the held drugs.

Dialysis disequilibrium syndrome

From a rapid reduction in electrolytes and other particles. Mild: nausea, vomiting, headache, fatigue, restlessness. Severe: mental status changes, seizures, coma.

Reducing the blood flow at the onset of symptoms can prevent it.

Peritoneal dialysis

Like you're 10

Here the filter is the lining of the client's own belly. Cleaning fluid goes in through a tube, sits for a while soaking up waste, and drains back out.

Peritonitis: the most common complication

Cloudy or opaque effluent is the earliest sign. Also fever, abdominal pain or tenderness, malaise, nausea, vomiting.

Nursing management, in order:

1. **Examine all effluent** for color and clarity. Normal is clear and light yellow.
2. If cloudy, **send a specimen** of the outflow for culture and sensitivity, Gram stain, and cell count.
3. Assess for fever, abdominal pain and tenderness, nausea, and vomiting.
4. Teach the client to give intraperitoneal antibiotics as prescribed and to complete the full course.

Other colors: brown means bowel perforation. The same color as urine, with the same glucose level, means bladder perforation.

Catheter care: **mask yourself and the client**, wash hands, sterile gloves, aseptic technique.

Kidney transplant

	Hyperacute	Acute	Chronic
Onset	Within 48 hours	1 week to any time after; over days to weeks	Gradual, months to years
Findings	Increased temperature and blood pressure; pain at the transplant site	Oliguria or anuria; temperature above 100 F; increased blood pressure; enlarged tender kidney; lethargy; rising creatinine, blood urea nitrogen, potassium; fluid retention	Gradual rise in blood urea nitrogen and creatinine; fluid retention; electrolyte changes; fatigue
Treatment	Immediate removal of the kidney	Increased immunosuppressive doses	Conservative until dialysis is required

Client education

Adherence to the antirejection drug regimen is what prevents rejection. Teach the signs of rejection and infection, and home monitoring.

PART 7 | HOW IT ALL CONNECTS

This is the section to reread on Sunday night. Each thread below ties a fact from one chapter to a fact in another. The individual facts come from the chapters; lining them up is the study aid.

Thread 1: Potassium

Like you're 10

Potassium belongs **inside** cells. Only a little should be in the blood. Aldosterone and the kidney push the extra out. Insulin pushes it into cells. Too much or too little in the blood, and the heart's rhythm goes wrong.

Condition	Potassium	Why
Adrenal insufficiency	High	No aldosterone to excrete it
Chronic kidney disease, acute kidney injury	High	The kidney cannot excrete it
Diabetic ketoacidosis, before treatment	Often mildly to moderately high	Common with hyperglycemia
Diabetic ketoacidosis, during treatment	Falls	Insulin, fluids, and correction of acidosis
Hyperaldosteronism, Cushing syndrome	Low	Too much mineralocorticoid effect
Spironolactone; ACE inhibitors	Risk of high	Potassium is retained

Pearl

IV insulin with dextrose shifts potassium into cells. It is a treatment in adrenal crisis hyperkalemia, and it is the reason potassium must be watched and replaced during diabetic ketoacidosis treatment. Same mechanism, opposite clinical concern.

Thread 2: Sodium and water

Condition	Sodium	Volume	Urine
Diabetes insipidus	High	Depleted	Huge volume, dilute
Syndrome of inappropriate antidiuretic hormone	Low (diluted)	Overloaded, usually no dependent edema	Low volume, concentrated
Adrenal insufficiency	Low (lost)	Depleted, hypotension	
Cushing syndrome, hyperaldosteronism	High	Overloaded, hypertension	
Myxedema coma	Low		
Late chronic kidney disease	Retained; level may look normal or falsely low from dilution	Overloaded, hypertension, edema	Low volume

Pearl: low sodium has two different stories

In adrenal insufficiency, sodium is **lost** and the client is dry and hypotensive: give saline. In syndrome of inappropriate antidiuretic hormone, sodium is **diluted** and the client is fluid overloaded: restrict fluid. Read the volume status before you pick the intervention.

Thread 3: Glucose

Glucose goes UP	Glucose goes DOWN
Diabetes mellitus	Too much insulin, sulfonylurea, or meglitinide
Cushing syndrome and glucocorticoid drugs	Adrenal insufficiency (no cortisol)
Acromegaly (growth hormone)	Myxedema coma
Hyperthyroidism	Chronic kidney disease in a client on insulin (insulin lasts longer)
Stress, infection, surgery	Alcohol, missed meals, exercise

Connection

Glucocorticoids are on the list of drugs that can contribute to hyperglycemic-hyperosmolar state. A client with type 2 diabetes started on prednisone needs closer glucose monitoring. A client with diabetes whose kidneys are failing needs **less** insulin, not more.

Thread 4: Calcium, phosphorus, and bone

- Parathyroid hormone and phosphorus move in **opposite directions** to calcium in parathyroid disease: hypoparathyroidism is calcium low and phosphorus high; hyperparathyroidism is calcium high and phosphorus low.
- Thyroidectomy** can injure the parathyroids: tingling around the mouth is hypocalcemia until proven otherwise.
- Chronic kidney disease**: phosphorus high, calcium low, parathyroid hormone high, bone loss.
- Cushing syndrome and hyperparathyroidism** both thin the bones: lift sheet, fall precautions.

Thread 5: Acid-base

Condition	Imbalance	Clue
Diabetic ketoacidosis	Metabolic acidosis	Ketones; Kussmaul respirations; fruity breath
Chronic kidney disease	Metabolic acidosis	Low bicarbonate; Kussmaul respirations
Adrenal insufficiency	Acidosis	Potassium retention promotes hydrogen ion reabsorption
Hyperaldosteronism	Metabolic alkalosis	Hydrogen ions lost with potassium
Hyperglycemic-hyperosmolar state	None	pH above 7.4; ketones negative

Pearl

Kussmaul respirations are the lungs compensating for metabolic acidosis. You will see them in **both** diabetic ketoacidosis and worsening kidney disease. Do not assume diabetes; check the ketones and the kidney function.

Thread 6: Diabetes and the kidney

- Diabetes is the leading cause of end-stage kidney disease. Hypertension speeds it up.
- Protect the kidney: glucose control, blood pressure control, ACE inhibitor or angiotensin receptor blocker for albuminuria, avoid NSAIDs and other nephrotoxins.
- Metformin is held around contrast media. SGLT2 inhibitors work at the kidney and raise the risk for dehydration and urinary tract infection.
- Diabetes raises the risk for acute kidney injury in any hospitalized client.
- Autonomic neuropathy causes urine retention, which leads to infection and kidney problems.

Thread 7: A stressor turns a chronic disease into a crisis

Infection, surgery, and trauma are named as triggers for every emergency in these chapters. When a stem gives a client with a known endocrine disease plus a new infection, think crisis.

Crisis	Hallmark findings	First-line actions
Acute adrenal insufficiency	Severe hypotension; sodium low; potassium high; hypoglycemia	Rapid saline; IV hydrocortisone; treat hyperkalemia and hypoglycemia; cardiac monitoring
Thyroid storm	High fever; tachycardia; agitation to coma	Airway; antithyroid drug; iodide; propranolol; cooling; nonsalicylate antipyretic
Myxedema coma	Hypothermia; bradycardia; hypoventilation; hypotension; hyponatremia; hypoglycemia	Airway; IV thyroid hormone; fluids; glucose; corticosteroids; warm blankets
Diabetic ketoacidosis	Glucose above 300; ketones; pH below 7.35; Kussmaul respirations	Fluids; IV regular insulin; potassium; dextrose added at 250 mg/dL
Hyperglycemic-hyperosmolar state	Glucose above 600; osmolarity above 320; no ketones; neurologic changes	Fluids to restore volume; hourly neurologic checks
Severe hypoglycemia	Cannot swallow; unconscious; seizures	Glucagon or IV dextrose; side-lying position
Hyperkalemia in kidney failure	Tall peaked T waves; dysrhythmias	Cardiac monitoring; kidney replacement therapy if above 6.5

Thread 8: The steroid thread

- Long-term glucocorticoids are the most common cause of **Cushing syndrome**.
- Stopping them suddenly is the most common cause of **secondary adrenal insufficiency** and can cause shock. Taper.
- Stress, illness, or surgery in a client on replacement requires a **higher** dose.
- Glucocorticoids are part of the treatment for thyroid storm, myxedema coma, and severe Graves eye disease, and they are given around adrenalectomy and replaced for life when the whole pituitary is removed.

- Their costs follow the client everywhere: hyperglycemia, infection with masked signs, thin skin, bone loss, gastric ulcers.

Thread 9: Lifelong replacement and a medical alert bracelet

The chapters call for both in: hypopituitarism, diabetes insipidus on desmopressin, adrenal insufficiency, bilateral adrenalectomy, hypothyroidism (including after thyroidectomy or radioactive iodine), and hypoparathyroidism. Diabetes: medical alert identification and always carry a carbohydrate source.

Thread 10: Weight is fluid

1 kg (2.2 lb) = 1 L of fluid. Daily weight, same scale, same time, same clothing, is the monitoring tool in diabetes insipidus, syndrome of inappropriate antidiuretic hormone, Cushing syndrome, adrenal insufficiency, and kidney disease.

Thread 11: Polyuria has more than one cause

Cause	How to tell
Diabetes mellitus	High glucose pulls water into the urine; glucose is elevated
Diabetes insipidus	Dilute urine, specific gravity below 1.005; sodium high; glucose not the issue
Early chronic kidney disease	Dilute urine with a reduced filtration rate
Diuretic phase of acute kidney injury	Follows oliguria; can signal the start of recovery

PART 8 | SELF-TEST

Rules

Eighteen questions. Commit to an answer for every one before turning to the key. For select-all items, judge **every** option on its own; do not stop after two or three that look right.

1. A client is 6 hours post transsphenoidal hypophysectomy and reports, "I keep having to swallow." The drip pad shows clear drainage with a light yellow ring at the edge. Which action does the nurse take?
 - A. Ask the client to blow the nose gently to clear the drainage
 - B. Keep the head of the bed elevated and test the drainage for glucose
 - C. Encourage coughing every hour to keep the airway clear
 - D. Provide a toothbrush for oral care to moisten the mouth
2. A client taking desmopressin calls the clinic. Which report requires the nurse to direct the client to the emergency department?
 - A. "I am urinating about 2 liters a day now."
 - B. "My nose feels irritated after I use the spray."
 - C. "I gained 3 pounds since yesterday, my head is pounding, and I feel confused."
 - D. "I felt thirsty and urinated more just before my dose was due."
3. A client with small cell lung cancer has a serum sodium of 118 mEq/L, a bounding pulse, and a weight gain of 1.5 kg in 24 hours. Which actions does the nurse include? **Select all that apply.**
 - A. Maintain the prescribed fluid restriction
 - B. Use saline rather than water to irrigate the nasogastric tube
 - C. Check orientation every 2 hours
 - D. Encourage 3 L of oral water daily
 - E. Reduce environmental noise and lighting
 - F. Record intake, output, and daily weight
4. A client has taken prednisone for 2 years. Which finding is the priority for follow-up?
 - A. Temperature of 99.6 F with a baseline of 98.4 F
 - B. Round face and fat pad between the shoulders
 - C. Reddish-purple striae on the abdomen
 - D. Fasting blood glucose of 148 mg/dL
5. Four hours after a total thyroidectomy, the client develops a harsh, high-pitched respiratory sound and oxygen saturation falls from 97% to 89%. Which action does the nurse take first?
 - A. Check for tingling around the mouth
 - B. Assess the voice for hoarseness
 - C. Inspect behind the neck for blood
 - D. Initiate the Rapid Response Team
6. A client newly prescribed levothyroxine describes the daily routine. Which statement indicates a need for more teaching?
 - A. "I take it with a glass of water as soon as I wake up."
 - B. "I wait about 45 minutes before I eat breakfast."
 - C. "I take it together with my calcium supplement so I do not forget either one."
 - D. "I will call if I have chest pain or feel short of breath."

7. Matrix. For each finding, mark whether it is consistent with **thyroid storm** or **myxedema coma**.

Finding	Thyroid storm	Myxedema coma
High fever		
Bradycardia		
Hypoglycemia		
Tachycardia		
Hypothermia		
Hyponatremia		
Tremors and anxiety		

8. A client who received insulin lispro 40 minutes ago is shaky and irritable but alert and able to swallow. Blood glucose is 62 mg/dL. Which action does the nurse take?

- A. Give glucagon intramuscularly
- B. Give 15 g of carbohydrate and recheck the glucose in 15 minutes
- C. Give crackers with nut butter and recheck in 1 hour
- D. Hold the next insulin dose and reassess at the next meal

9. A client with diabetic ketoacidosis is receiving 0.9% saline and a continuous regular insulin infusion. The blood glucose is now 248 mg/dL. Which prescription does the nurse anticipate?

- A. Change the IV fluid to 5% dextrose in 0.45% saline
- B. Discontinue the insulin infusion
- C. Give IV sodium bicarbonate
- D. Change to subcutaneous insulin while the client remains NPO

10. The same client's potassium has fallen from 5.6 to 4.7 mEq/L and IV potassium is prescribed. Which finding must the nurse confirm before starting it?

- A. Blood glucose below 200 mg/dL
- B. Urine ketones negative
- C. Serum bicarbonate above 18 mEq/L
- D. Urine output of at least 30 mL/hr

11. An 80-year-old client with type 2 diabetes is brought in confused after 5 days of poor fluid intake. Glucose 940 mg/dL, serum osmolarity 348 mOsm/L, pH 7.42, serum ketones negative. Which is the first priority of care?

- A. Giving sodium bicarbonate to correct acidosis
- B. Starting subcutaneous insulin before any fluids
- C. Replacing fluid to increase blood volume
- D. Restricting fluids to prevent cerebral edema

12. A client with type 2 diabetes who takes metformin is scheduled for a CT scan with contrast. The estimated glomerular filtration rate is 52. Which actions does the nurse anticipate? **Select all that apply.**

- A. Holding metformin for at least 24 hours before and after the procedure
- B. Confirming that the radiologist is aware of the reduced kidney function
- C. Giving IV fluids before and after the procedure
- D. Restricting fluids for 6 hours after the procedure
- E. Monitoring urine output after the procedure

13. A client with stage 4 chronic kidney disease has new prescriptions. Which one does the nurse question?

- A. Calcium acetate with meals
- B. Magnesium hydroxide antacid as needed
- C. Epoetin alfa subcutaneously
- D. Furosemide daily

14. A client has a mature arteriovenous fistula in the left forearm. Which action by assistive personnel requires the nurse to intervene?

- A. Placing the blood pressure cuff on the left arm
- B. Reporting oozing at a needle site
- C. Reminding the client not to sleep on the left arm
- D. Encouraging the client to do range-of-motion exercises

15. A client on peritoneal dialysis reports that the fluid draining out looks cloudy. Which action does the nurse take?

- A. Reassure the client that cloudy fluid is expected once dialysis is established
- B. Prepare the client for repair of a bladder perforation
- C. Increase the dwell time of the next exchange
- D. Send a specimen of the outflow for culture and sensitivity, Gram stain, and cell count

16. A client with end-stage kidney disease has bone pain. Calcium is 7.9 mg/dL and phosphorus is 6.8 mg/dL. Which hormone does the nurse expect to be elevated?

- A. Calcitonin
- B. Aldosterone
- C. Parathyroid hormone
- D. Antidiuretic hormone

17. Matrix. Match each laboratory and fluid pattern to the condition it best fits. Each condition is used once: adrenal insufficiency, hyperaldosteronism, diabetes insipidus, syndrome of inappropriate antidiuretic hormone.

Pattern	Condition
Sodium low, potassium high, hypotension, salt craving	
Sodium high, potassium low, hypertension, metabolic alkalosis	
Sodium high, urine specific gravity 1.002, urine output 9 L per day	
Sodium low, bounding pulse, concentrated urine, weight gain without dependent edema	

18. Two hours into hemodialysis, a client reports headache and nausea and becomes restless. Which action is expected to prevent progression?

- A. Increasing the blood flow rate to finish sooner
- B. Reducing the blood flow rate
- C. Giving the held antihypertensive dose
- D. Taking the blood pressure in the access arm

ANSWER KEY WITH RATIONALES

1. **B.** Increased swallowing and a halo sign indicate a possible cerebrospinal fluid leak. The chapter directs the nurse to keep the head of the bed elevated and assess the drainage, including for glucose. Blowing the nose (A) and coughing (C) raise intracranial pressure and are prohibited. Toothbrushing (D) is avoided for about 2 weeks.
2. **C.** Weight gain of more than 2.2 lb with persistent headache and acute confusion signals water toxicity from desmopressin. A is improvement. B is a known effect of the intranasal form that calls for a route change, not an emergency. D means the next dose is needed.
3. **A, B, C, E, F.** This is syndrome of inappropriate antidiuretic hormone with dilutional hyponatremia. Fluid is restricted; saline replaces water for tube irrigation; sodium below 120 calls for a safe, low-stimulation environment with orientation checks every 2 hours; intake, output, and weight track the response. D adds free water and lowers sodium further.
4. **A.** Excess cortisol masks infection. A rise of even 1 F above baseline is significant and indicates infection until proven otherwise. B, C, and D are expected findings of hypercortisolism. Actual infection outranks expected chronic changes.
5. **D.** Stridor with falling saturation is airway obstruction. The chapter's direction is to initiate the Rapid Response Team immediately. A, B, and C are valid assessments for hypocalcemia, laryngeal nerve damage, and hemorrhage, but the airway is failing now; intervention precedes further assessment.
6. **C.** Calcium supplements, iron, and antacids reduce absorption and must be separated from levothyroxine by at least 4 hours. A, B, and D are correct practice.
7. **Thyroid storm:** high fever, tachycardia, tremors and anxiety. **Myxedema coma:** bradycardia, hypoglycemia, hypothermia, hyponatremia. Storm is everything too fast and hot; myxedema coma is everything too slow and cold.
8. **B.** The client is conscious and can swallow with a glucose below 70: the 15-15 rule applies. Glucagon (A) is for a client who cannot swallow. A complex carbohydrate (C) comes after the glucose has recovered, and 1 hour is too long. D ignores an actual problem.
9. **A.** When glucose reaches 250 mg/dL, 5% dextrose in 0.45% saline is started to prevent hypoglycemia and cerebral edema while insulin continues. The insulin is not stopped (B); ketosis is not yet resolved. Bicarbonate (C) is only for pH 7.0 or less. Subcutaneous insulin (D) starts when the client can take oral fluids and ketosis has stopped.
10. **D.** Urine output of at least 30 mL/hr must be confirmed before any IV potassium. A, B, and C are resolution criteria for diabetic ketoacidosis, not requirements for potassium.
11. **C.** Glucose above 600, osmolarity above 320, normal pH, and no ketones is hyperglycemic-hyperosmolar state. The first priority is to increase blood volume. There is no acidosis to correct (A). Fluids precede insulin (B). Restricting fluid (D) worsens the cause.
12. **A, B, C, E.** Metformin is held at least 24 hours before and after contrast. Kidney function is communicated to the radiologist so the lowest dose is used. IV fluids before and after dilute the contrast and speed its elimination, and a urine output target is monitored. D is the opposite of what is needed.
13. **B.** Clients with kidney disease cannot excrete magnesium and must avoid magnesium-containing antacids. Calcium acetate with meals (A) binds phosphorus. Epoetin alfa (C) treats anemia. Furosemide (D) is used while urine is still produced.
14. **A.** Blood pressure measurement, venipuncture, and IV lines are prohibited in the access arm because compression can destroy the access. B, C, and D are correct actions.
15. **D.** Cloudy or opaque effluent is the earliest sign of peritonitis; a specimen is sent for culture and sensitivity, Gram stain, and cell count. Established effluent should be clear and light yellow (A). Bladder perforation (B) gives urine-colored outflow. C does not address infection.

16. C. Failing kidneys retain phosphorus and cannot activate vitamin D; calcium falls; the parathyroid glands release more hormone, which pulls calcium from bone. This is renal osteodystrophy.

17. In order: adrenal insufficiency; hyperaldosteronism; diabetes insipidus; syndrome of inappropriate antidiuretic hormone. Rows 1 and 2 are aldosterone problems (too little, too much). Rows 3 and 4 are antidiuretic hormone problems (too little, too much).

18. B. These are mild signs of dialysis disequilibrium syndrome. Reducing the blood flow at the onset of symptoms can prevent it. A makes the shift faster. C risks hypotension. D is prohibited.

If you missed more than four

Sort the misses by thread in Part 7, not by chapter. Most errors on this material come from two places: mixing up the two antidiuretic hormone disorders, and choosing another assessment when the stem already describes an airway or perfusion failure that calls for action.